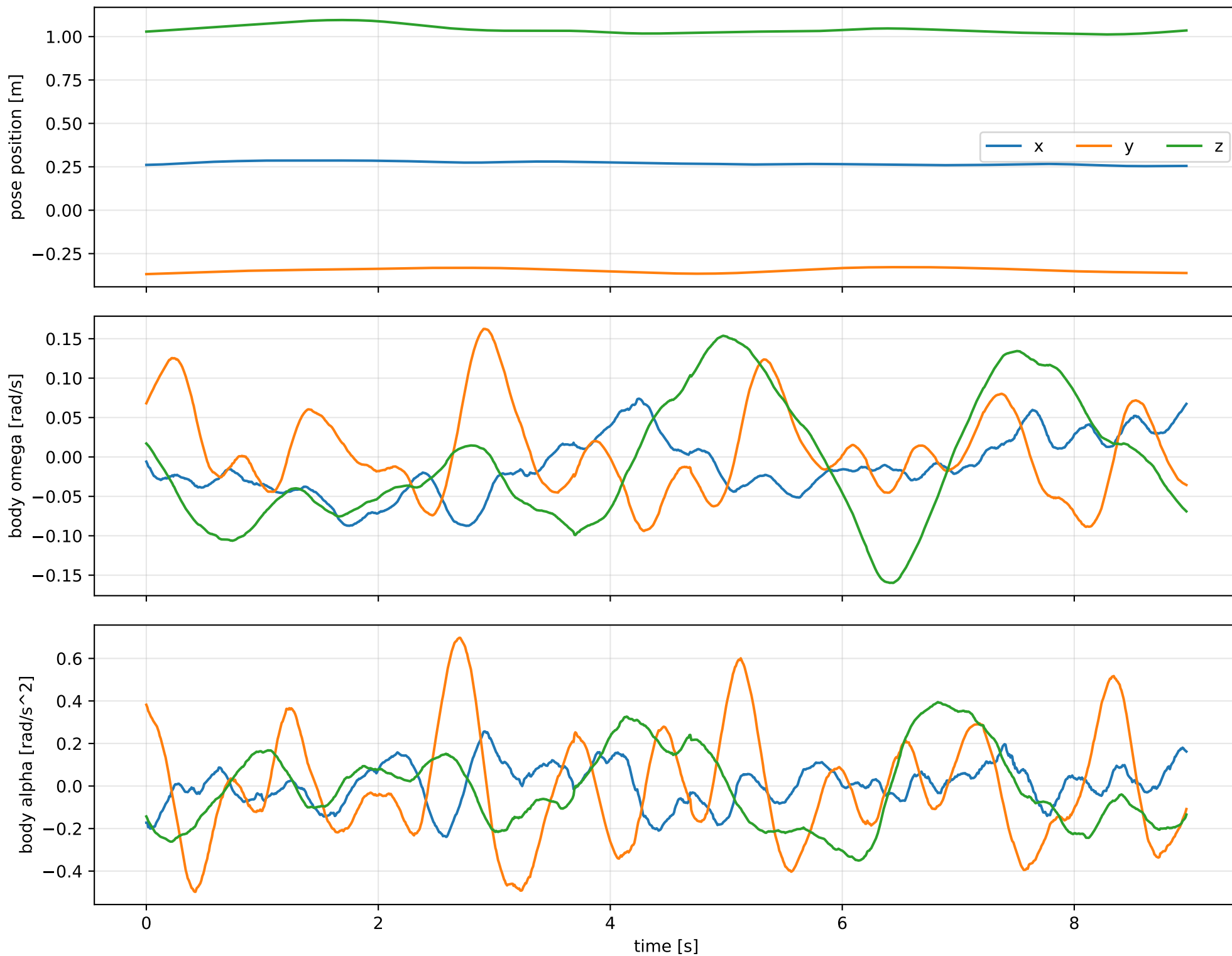
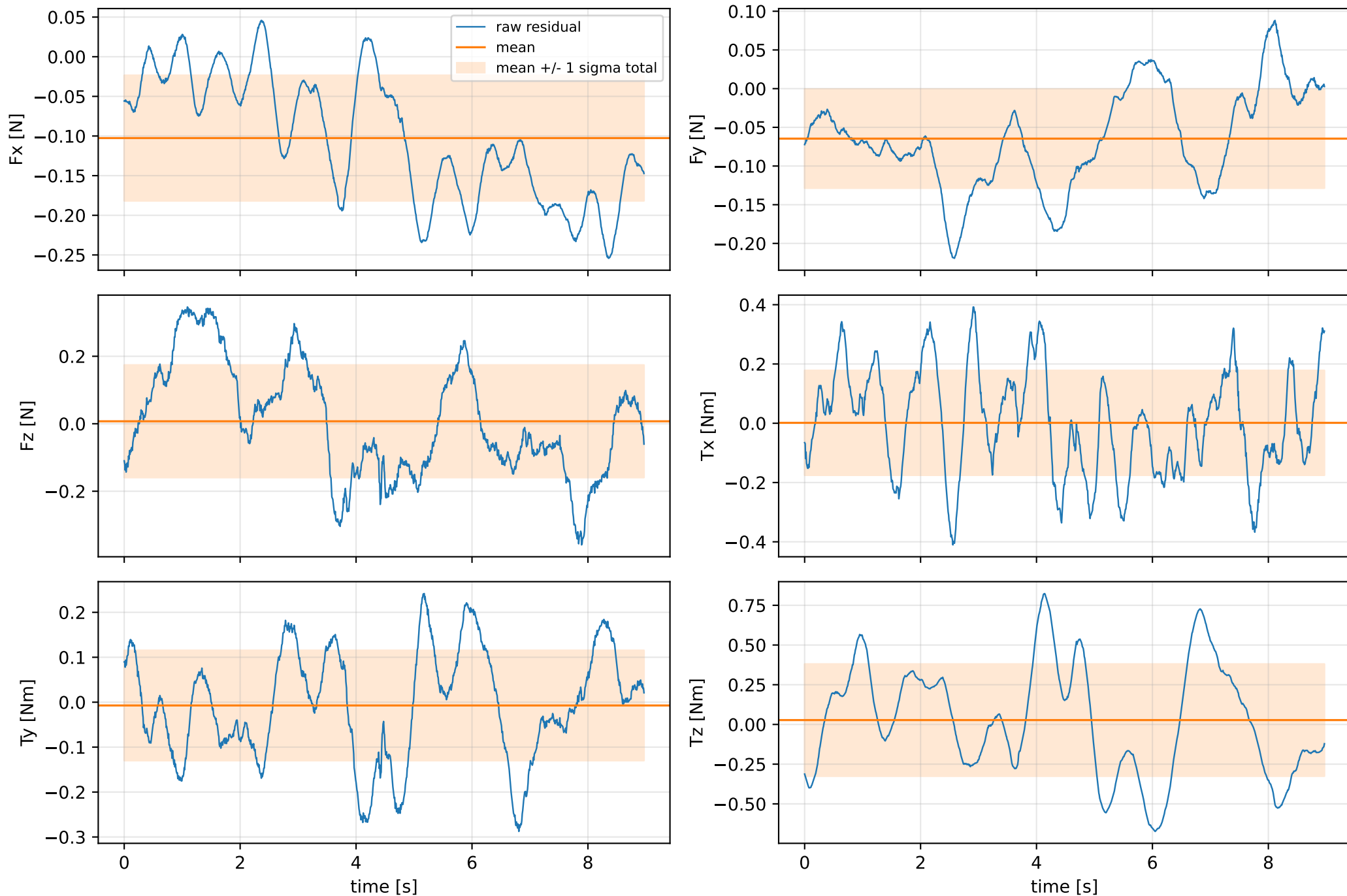


force_over_mass_rotor_4_nominal: SG trajectory and derived motion



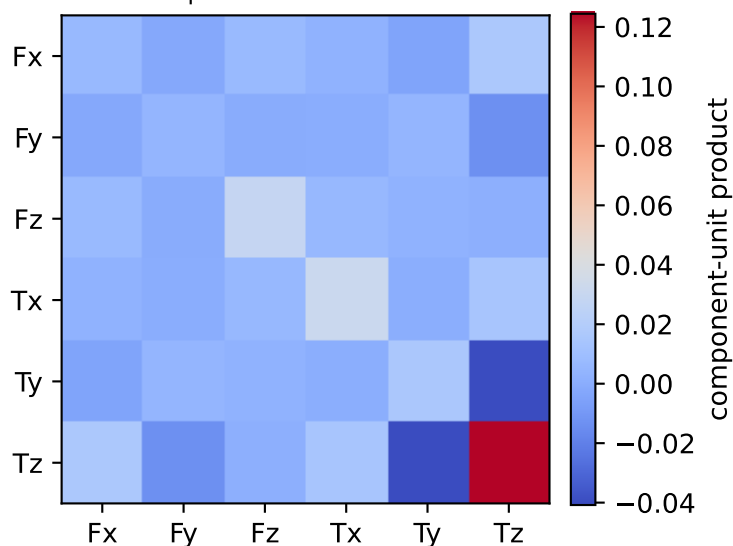
force_over_mass_rotor_4_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



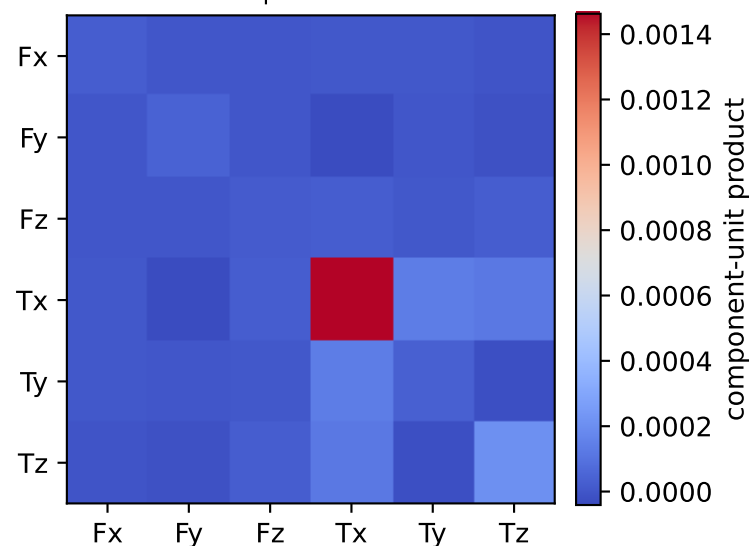
vector RMS about zero: force 0.22989 N, torque 0.41394 Nm; component std about mean: force [0.0791 0.064 0.1667], torque [0.1767 0.1226 0.3527]

force_over_mass_rotor_4_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0. 0.002 0.003 0.024 0.034 0.143]

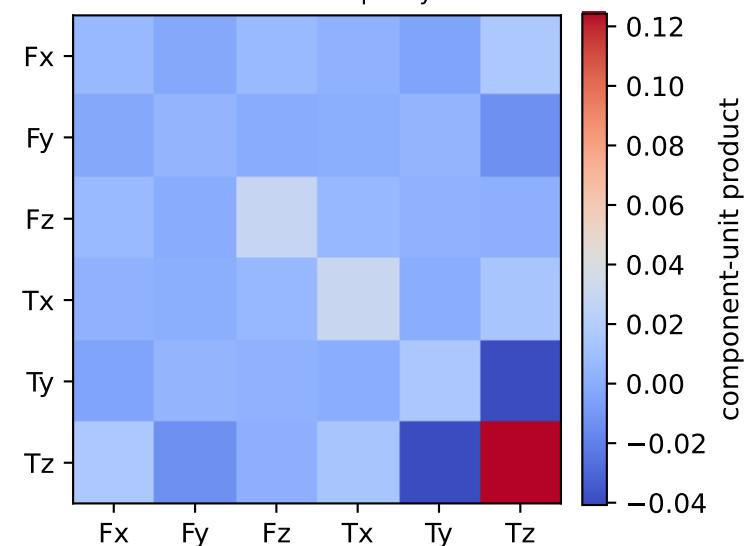
empirical total covariance



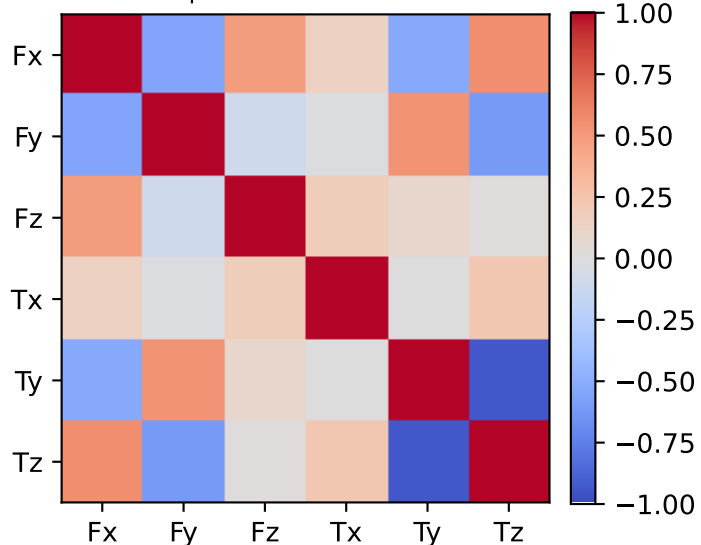
mean full-SG prediction covariance



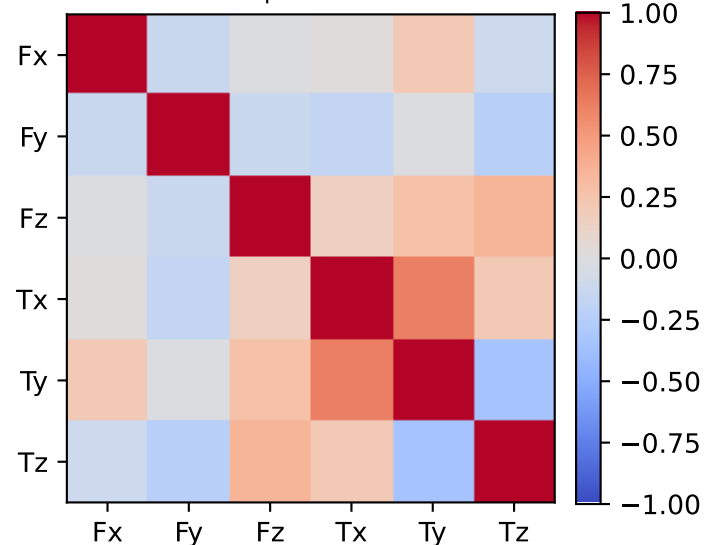
PSD excess model discrepancy covariance



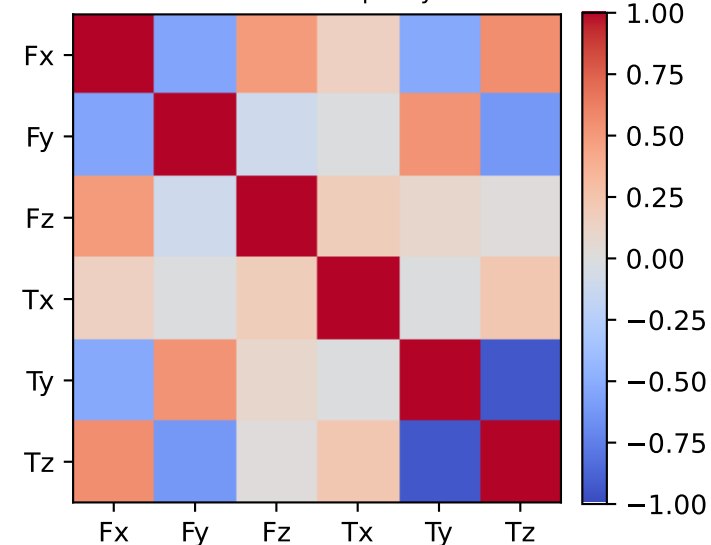
empirical total correlation



mean full-SG prediction correlation

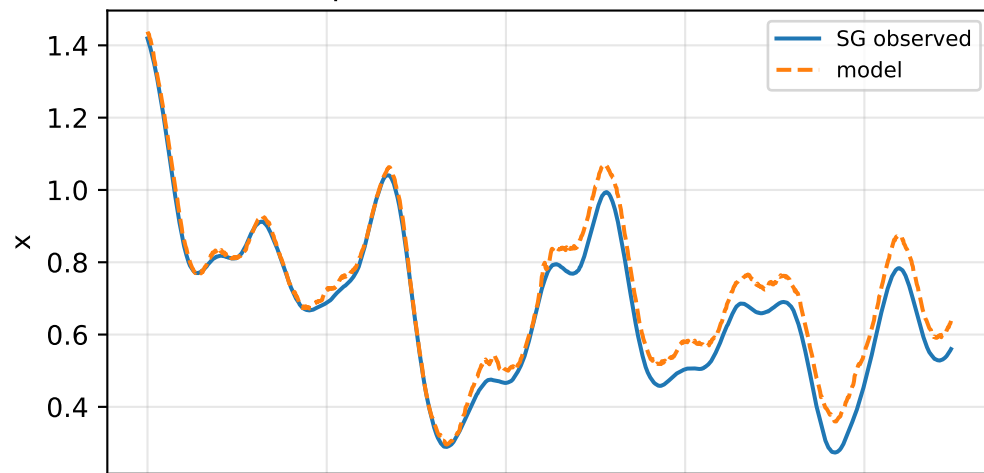


PSD excess model discrepancy correlation

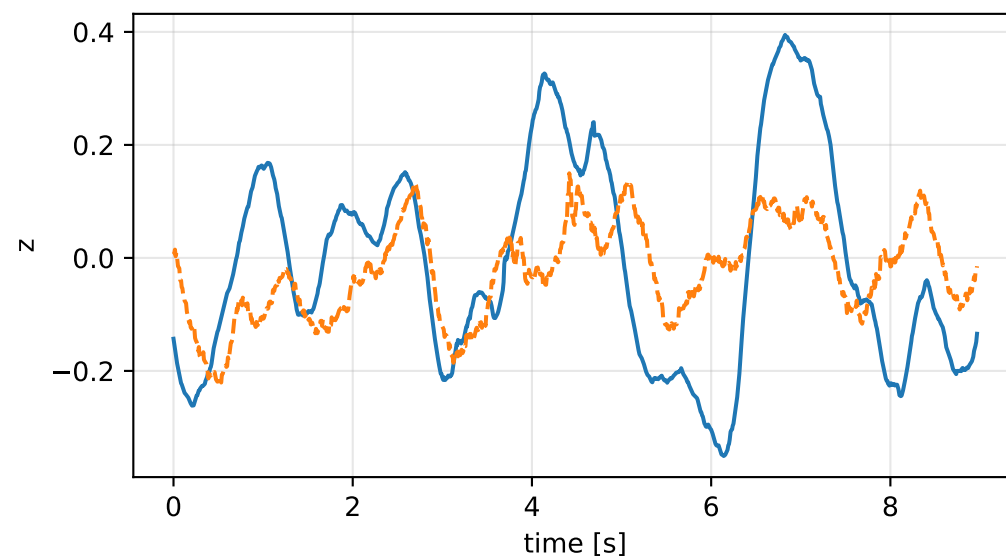
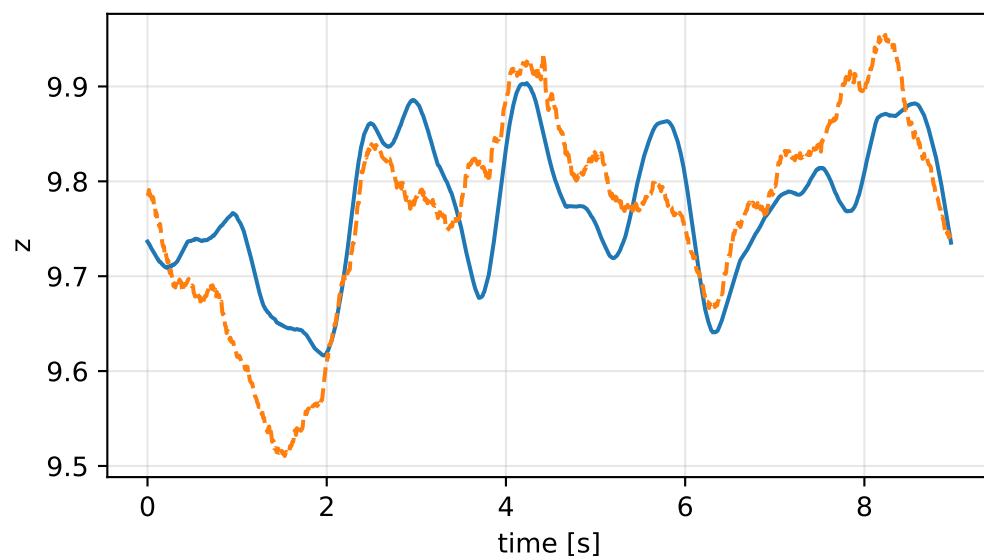
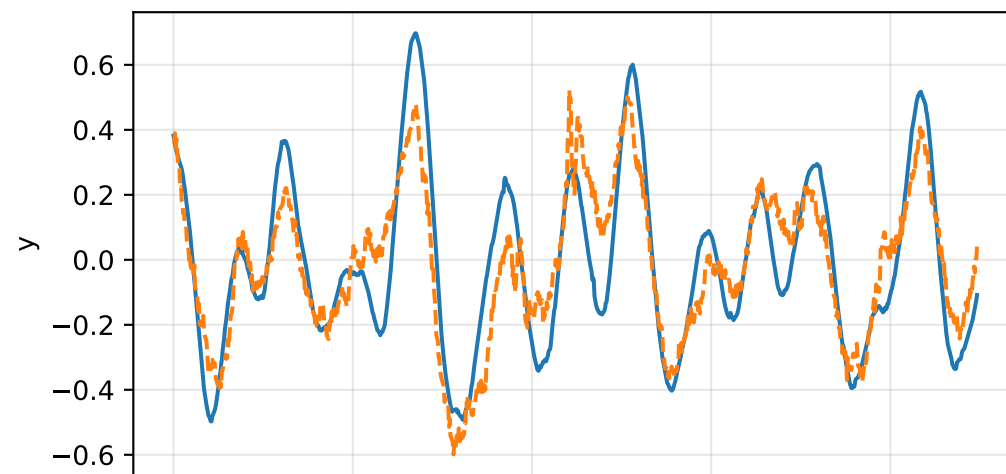
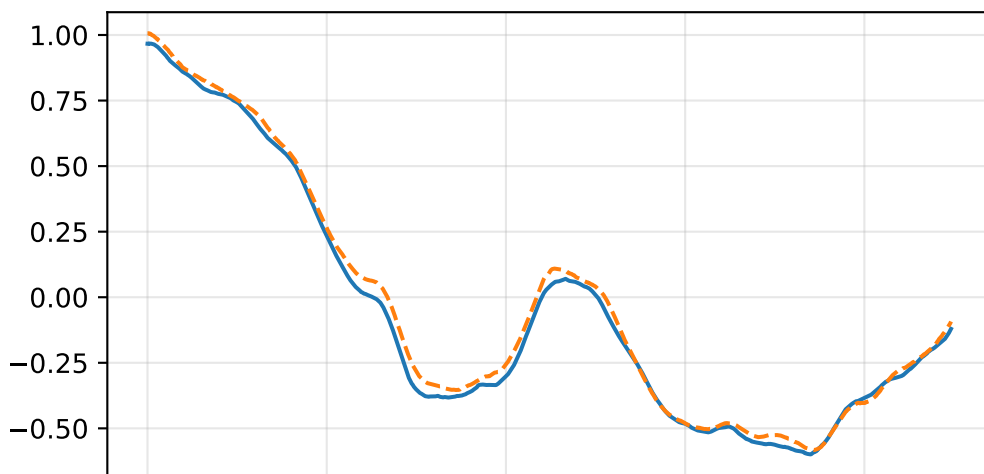
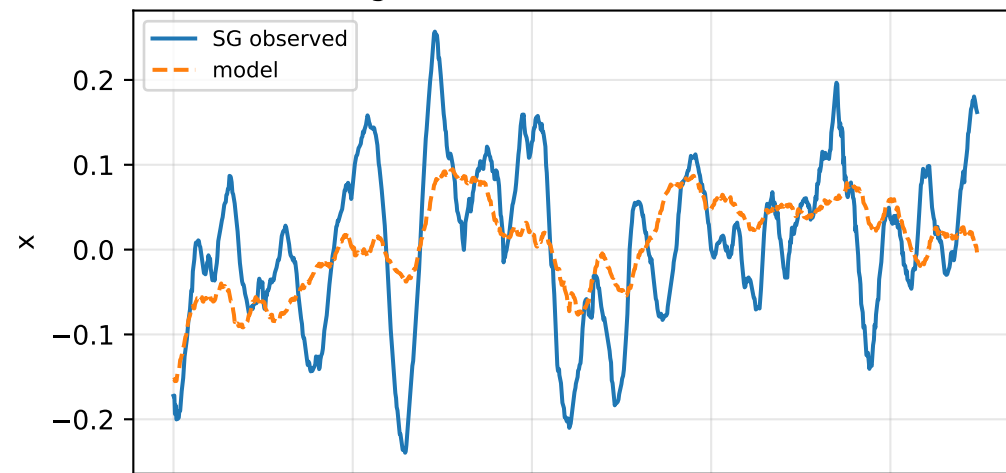


force_over_mass_rotor_4_nominal: acceleration residual objective

specific acceleration [m/s^2]

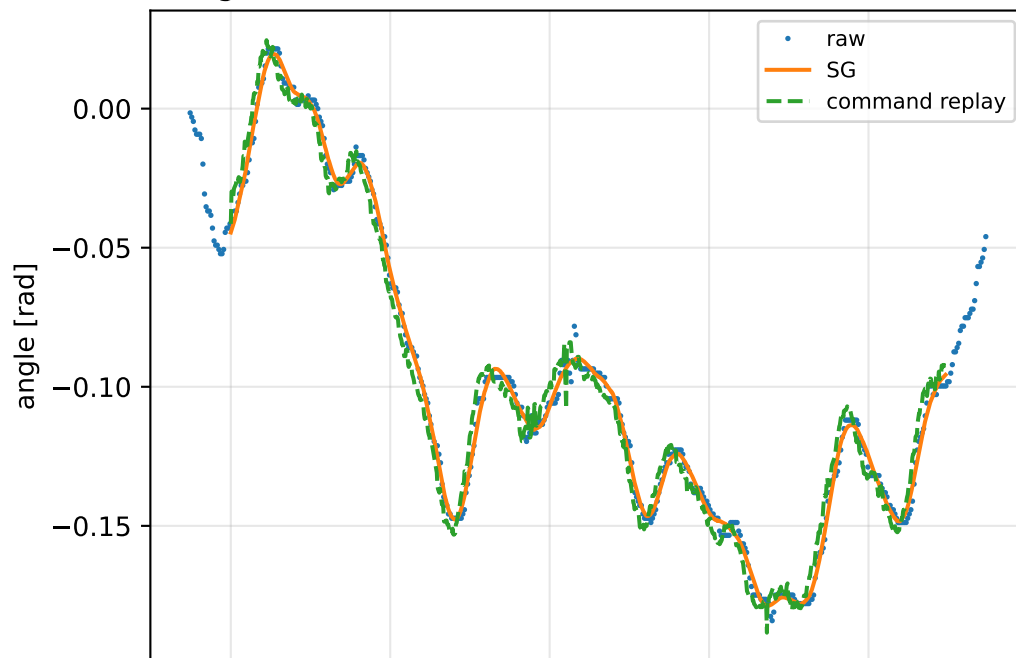


angular acceleration [rad/s^2]

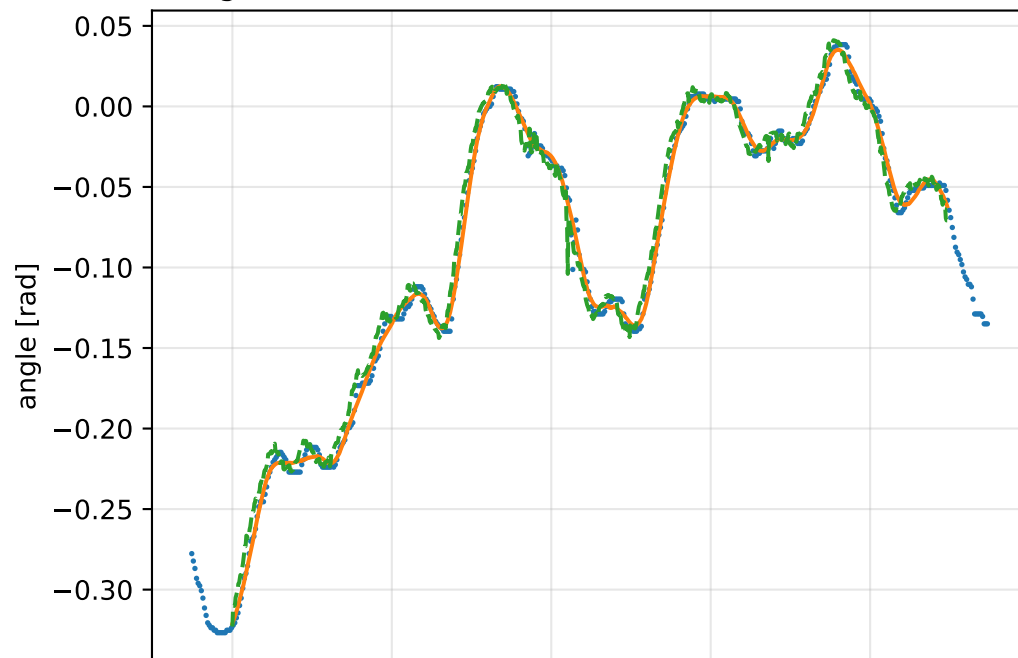


force_over_mass_rotor_4_nominal: gimbal measurement audit (objective=measured_sg)

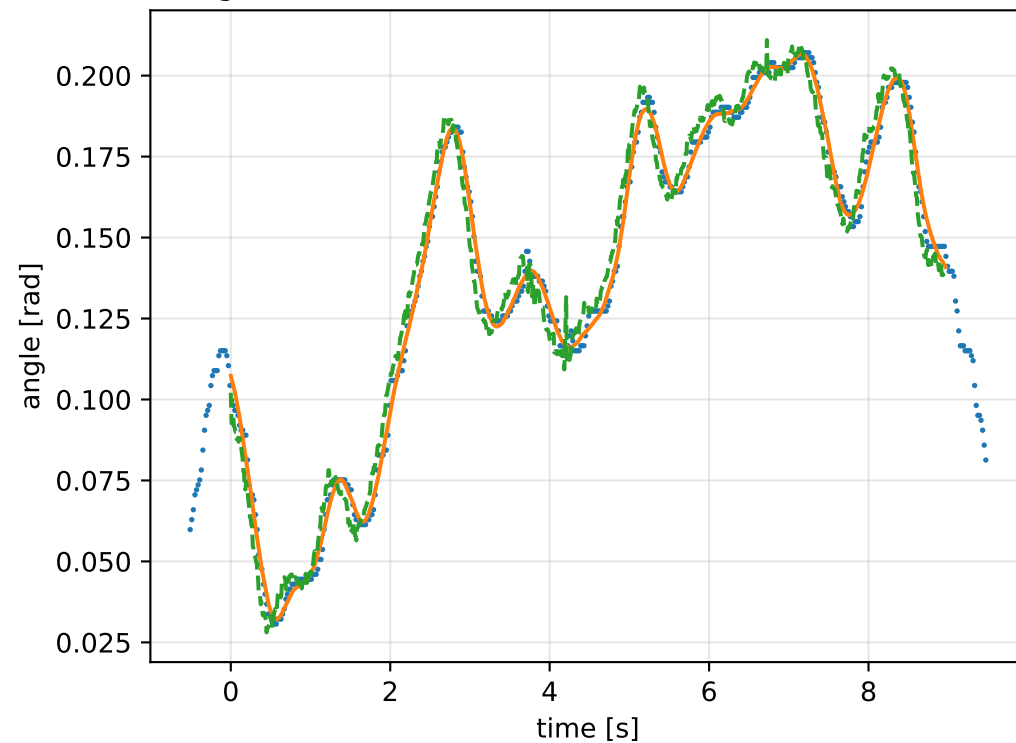
gimbal 1: RMSE=0.006345, max=0.01494 rad



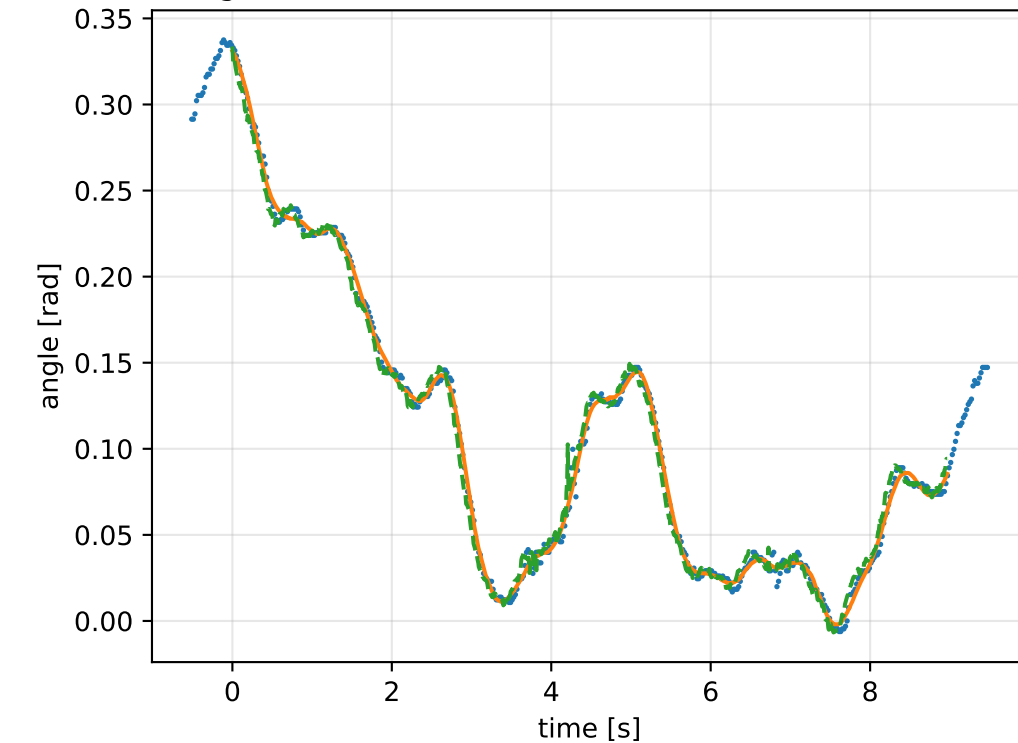
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

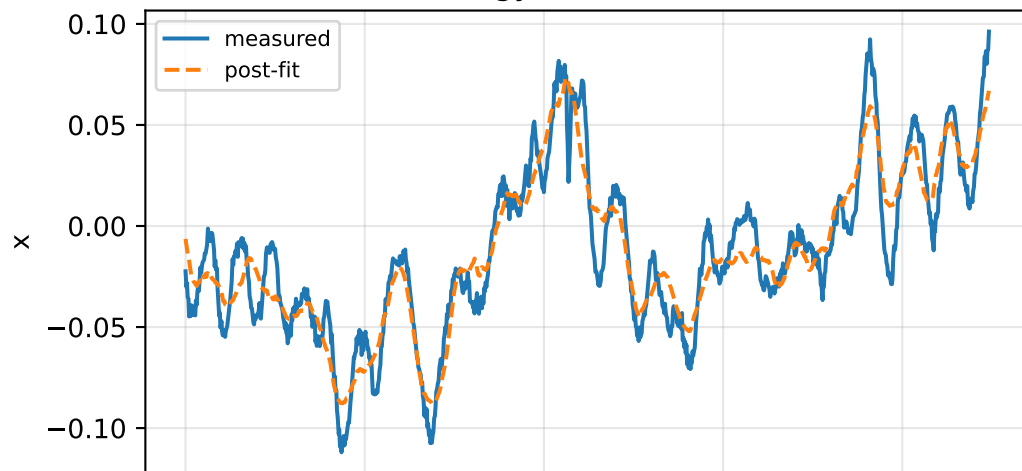


gimbal 4: RMSE=0.006321, max=0.03642 rad

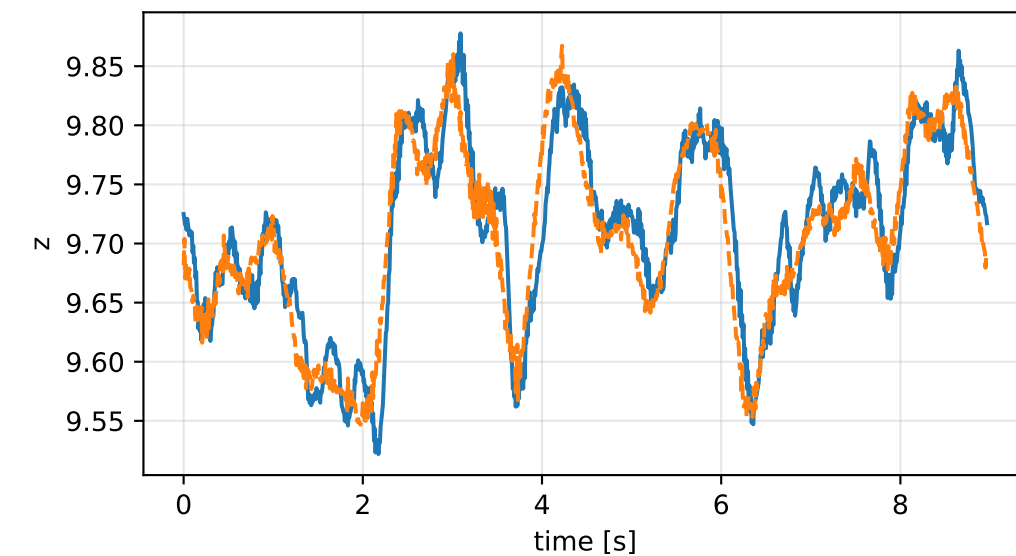
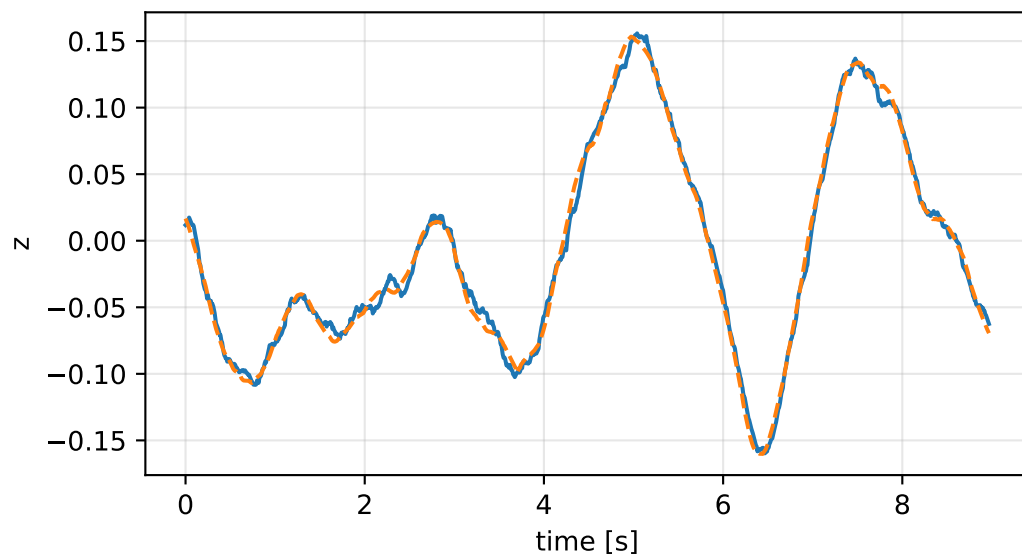
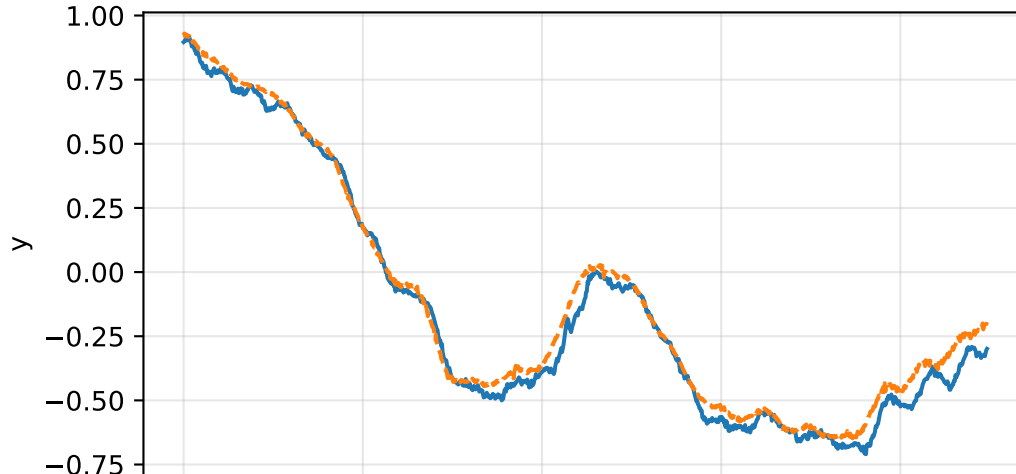
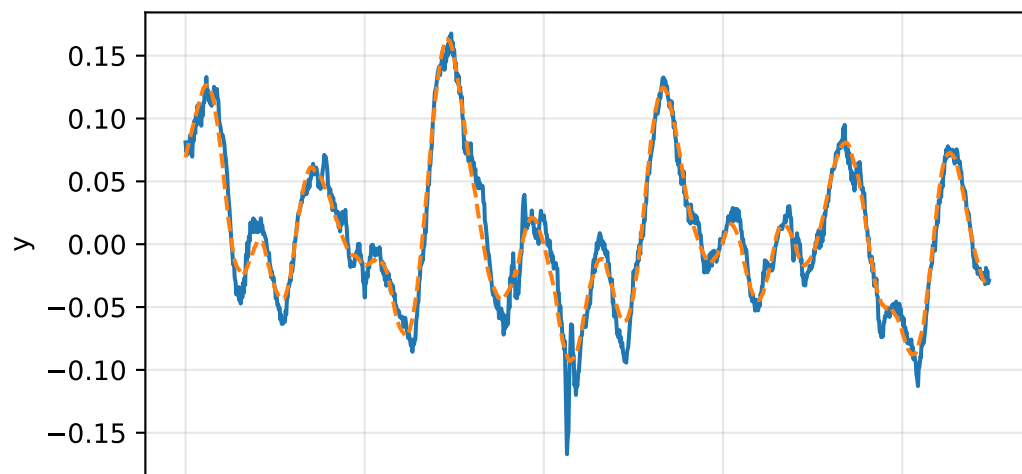
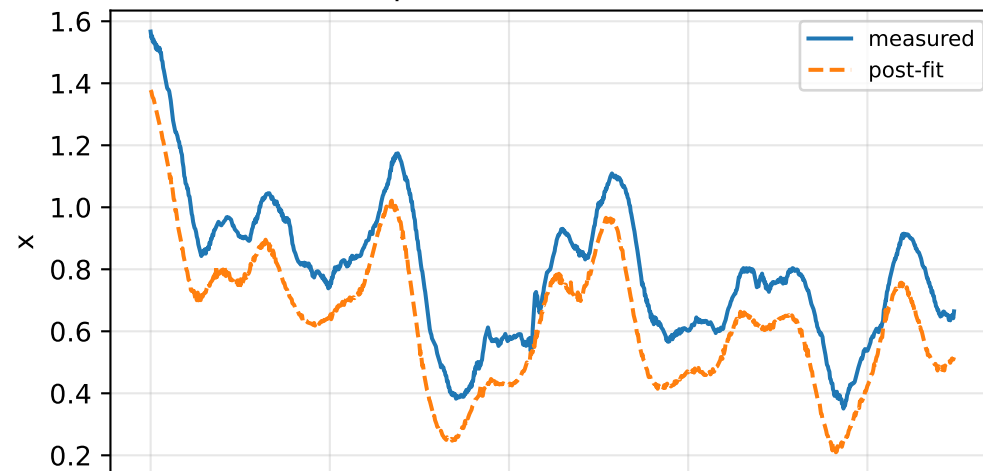


force_over_mass_rotor_4_nominal: IMU comparison (diagnostic only)

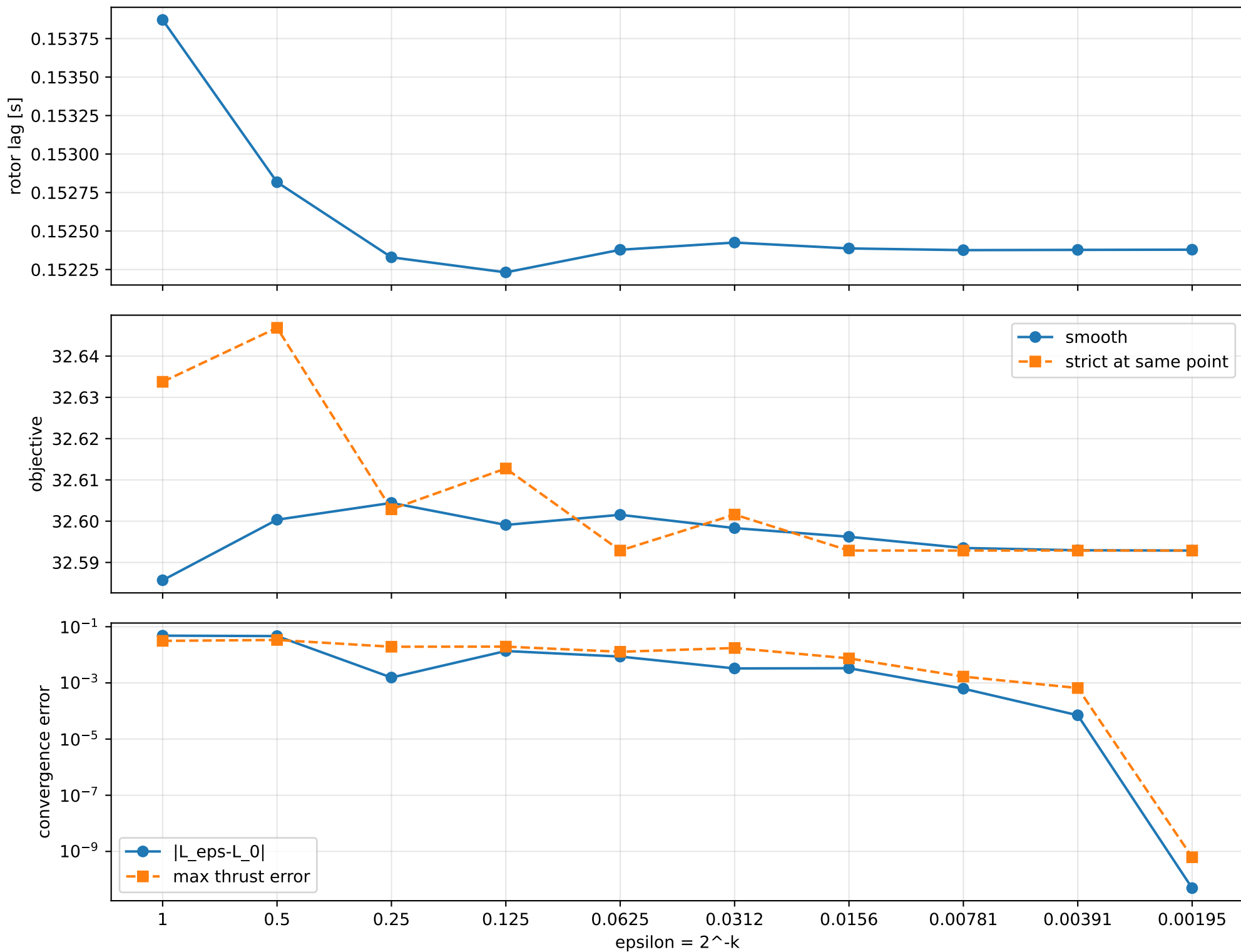
gyro [rad/s]



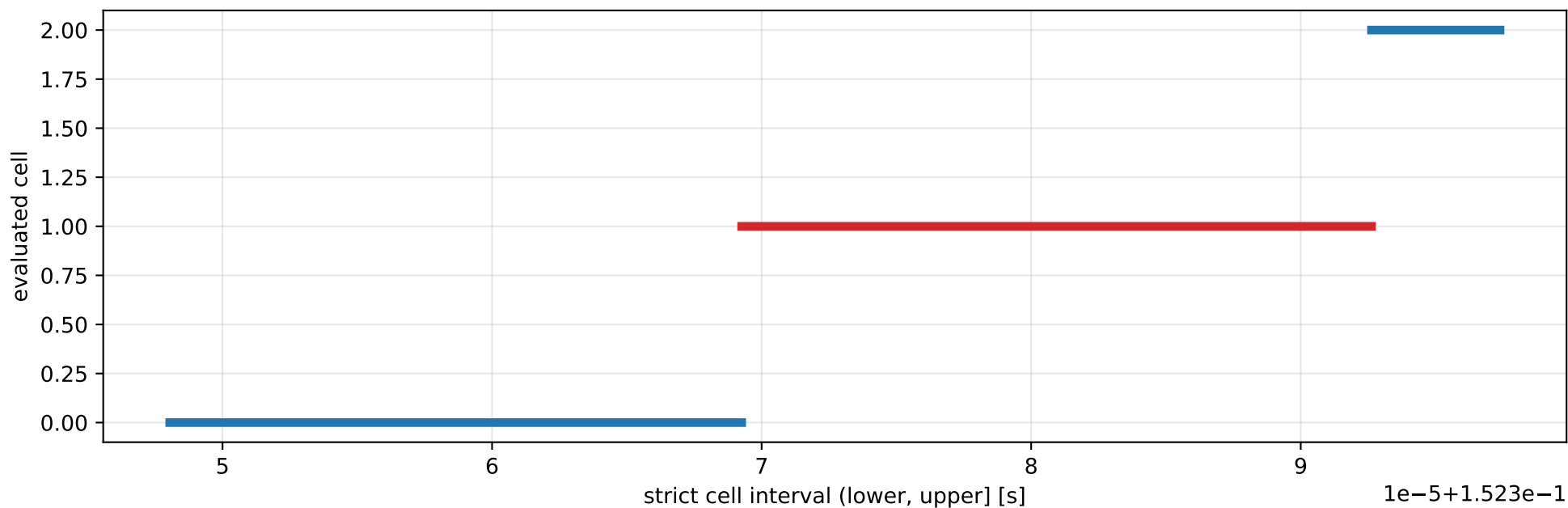
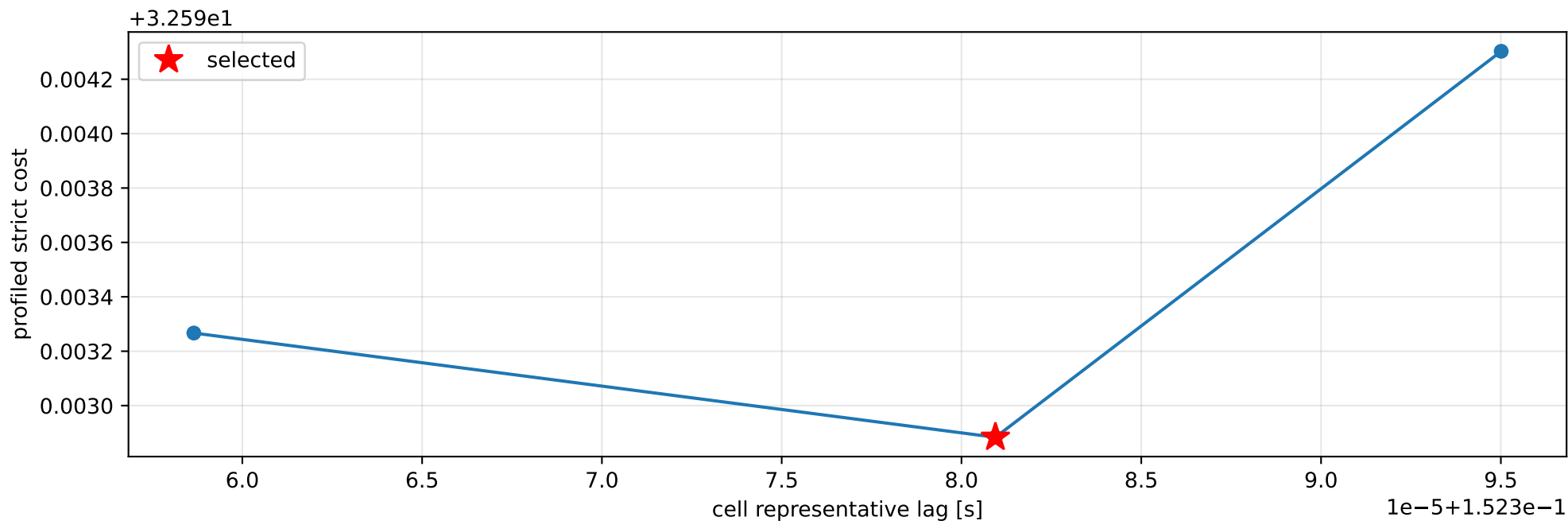
specific force [m/s^2]



force_over_mass_rotor_4_nominal: smooth-to-strict continuation



force_over_mass_rotor_4_nominal: exact strict-ZOH lag cells



selected (0.152369261, 0.152392626] s; final smooth 0.15237898 s; data boundary=False

force_over_mass_rotor_4_nominal: parameter estimate

Case: force_over_mass_rotor_4_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.09613424

inertia [kg m²]:

```
[[ 1.8957665  0.11906001  0.15877429]
 [ 0.11906001  0.40616562 -0.49723081]
 [ 0.15877429 -0.49723081  1.74429445]]
```

CoG body [m]: [-0.09242307 -0.02315427 -0.06492022]

force effectiveness: [1.04637037 0.56117176 0.35740492 0.89136604]

scale-free J/m [m²]:

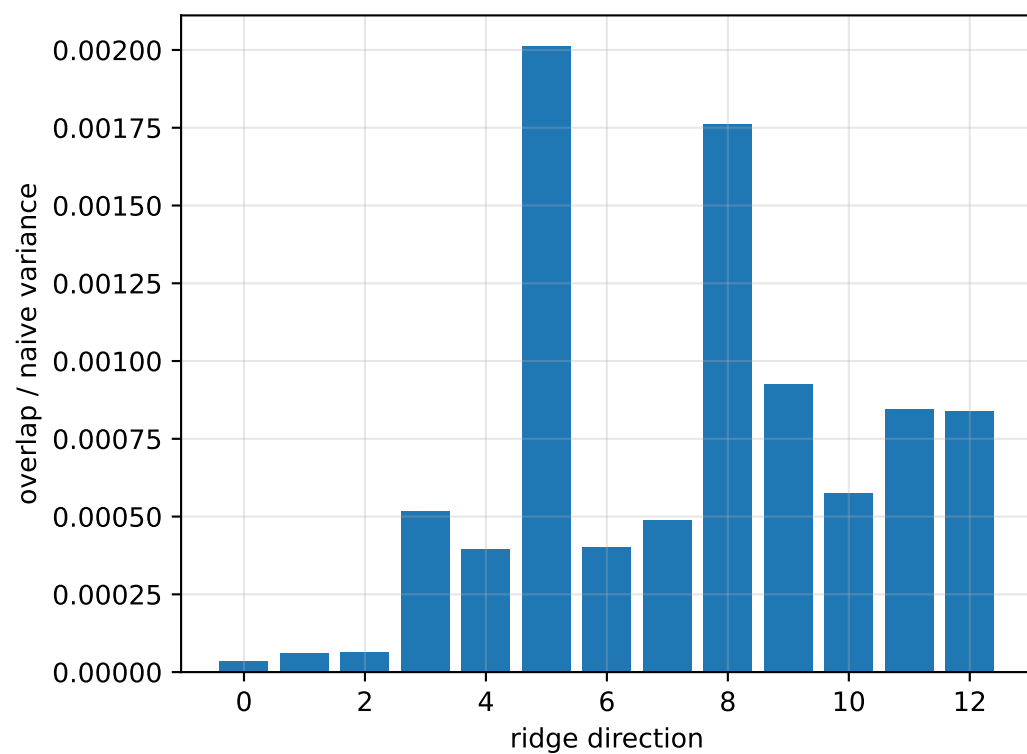
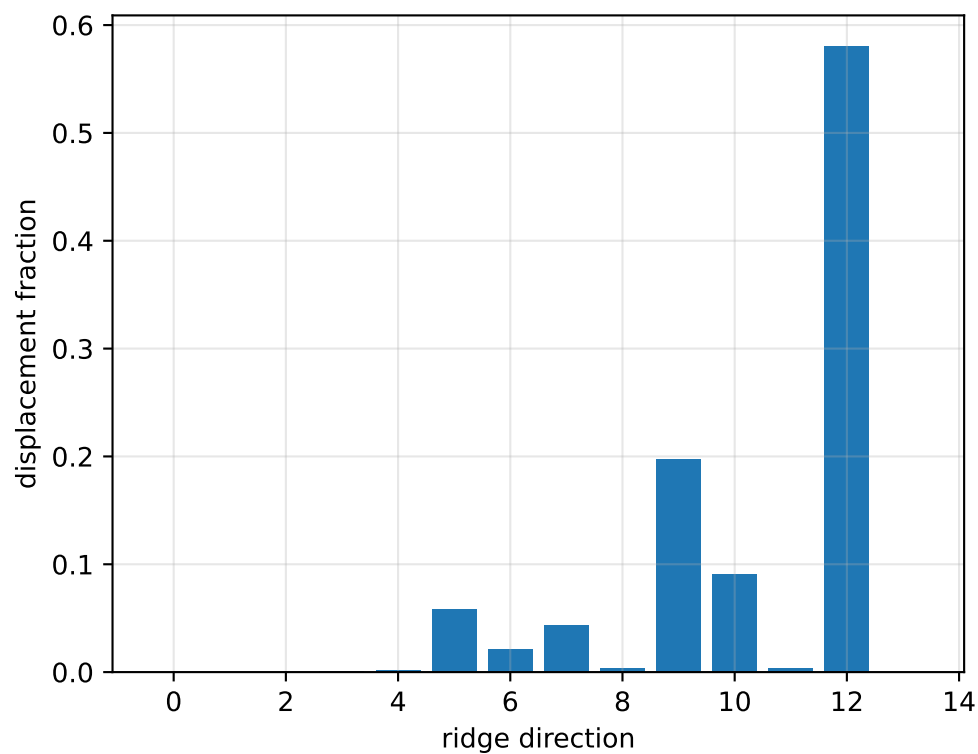
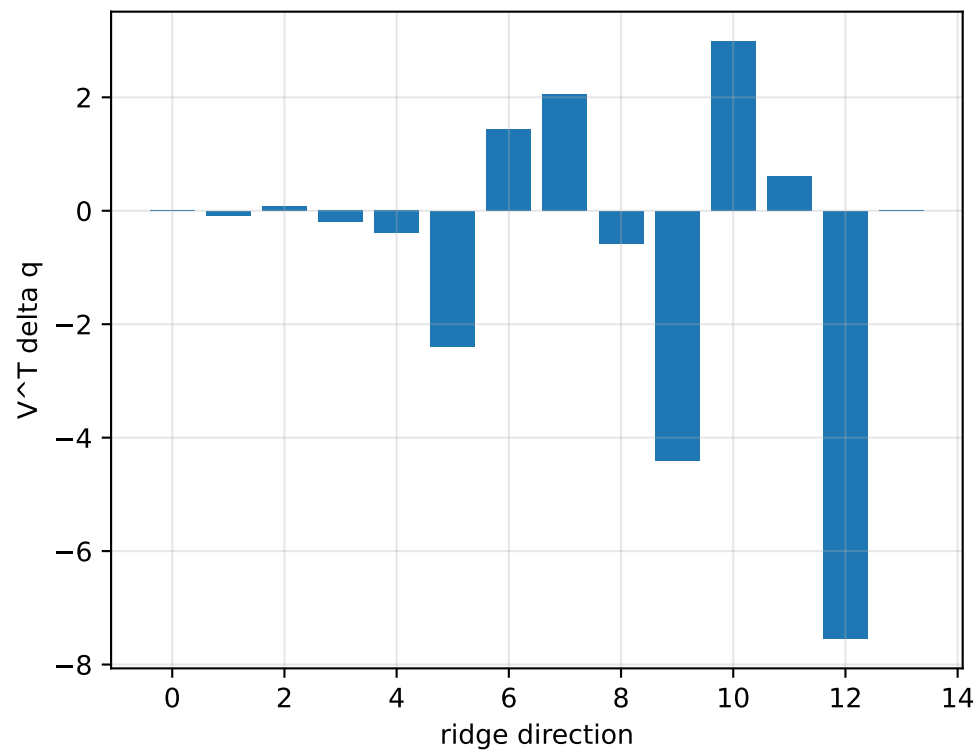
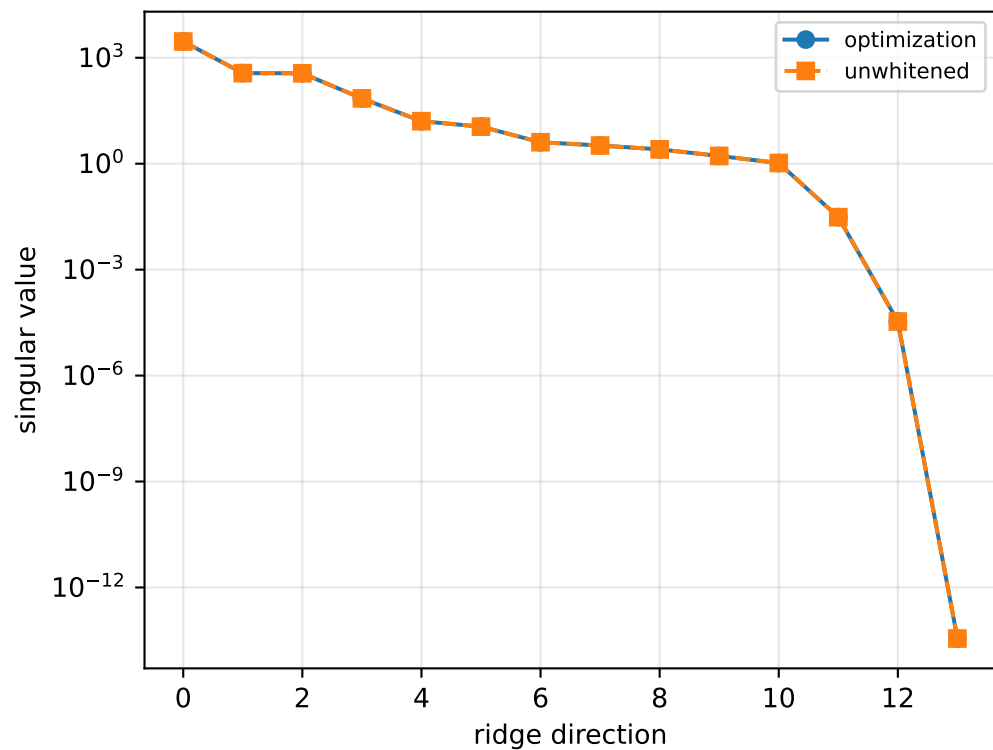
```
[[ 0.90441083  0.0567998  0.07574624]
 [ 0.0567998  0.1937689 -0.23721325]
 [ 0.07574624 -0.23721325  0.83214825]]
```

scale-free f/m: [0.49919053 0.26771747 0.1705067 0.42524282]

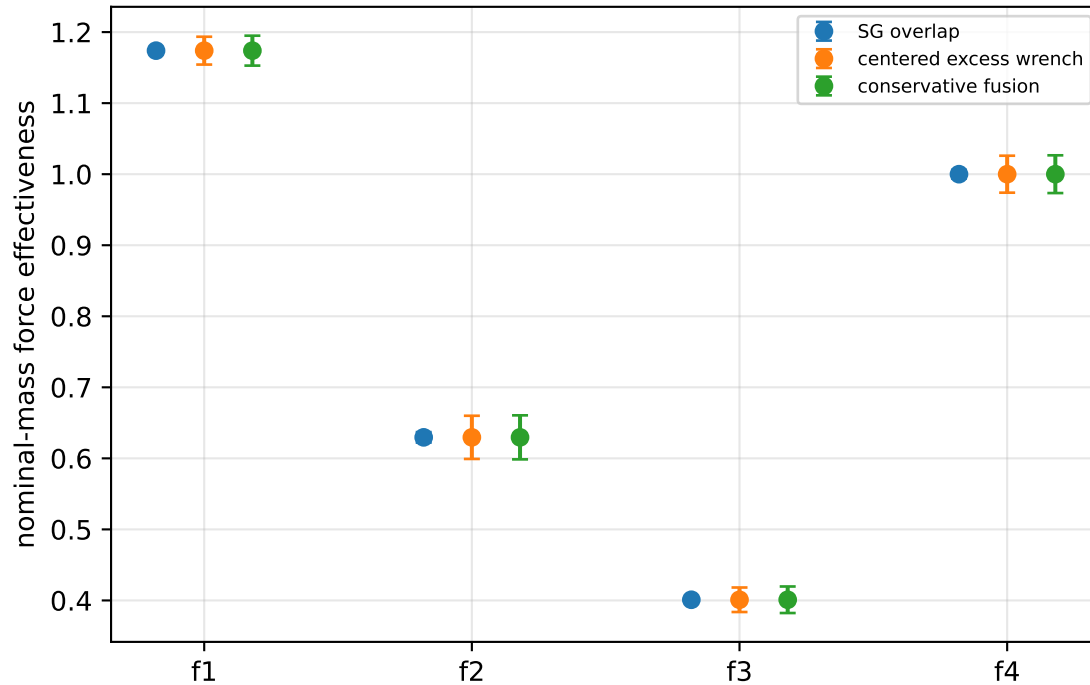
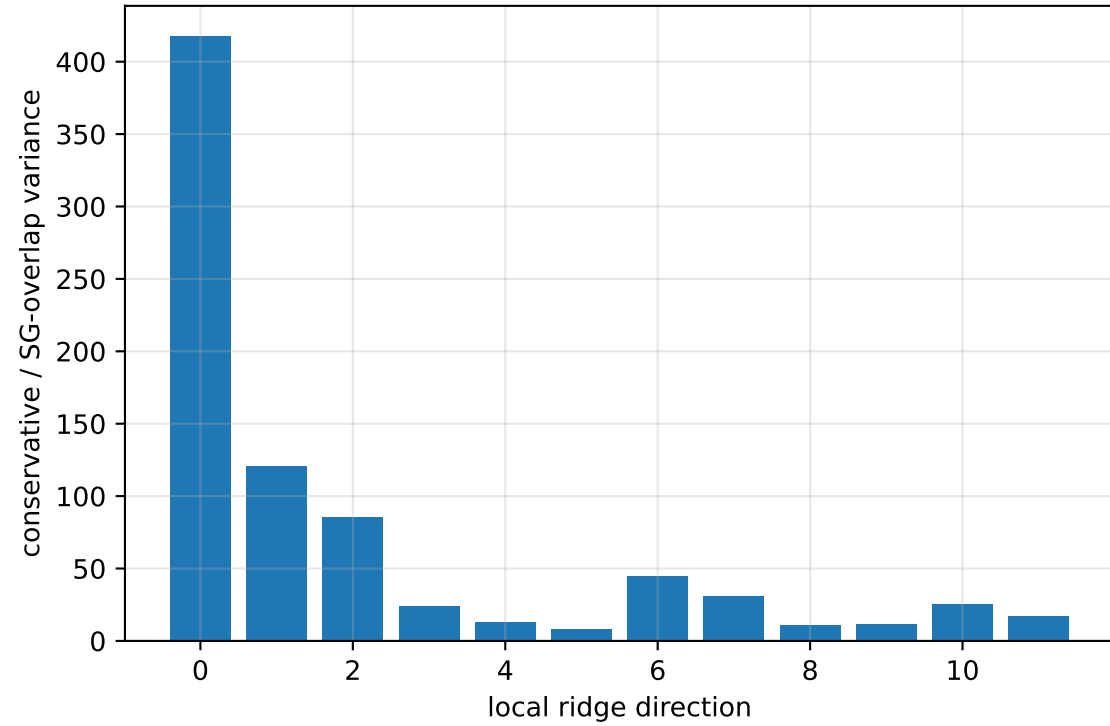
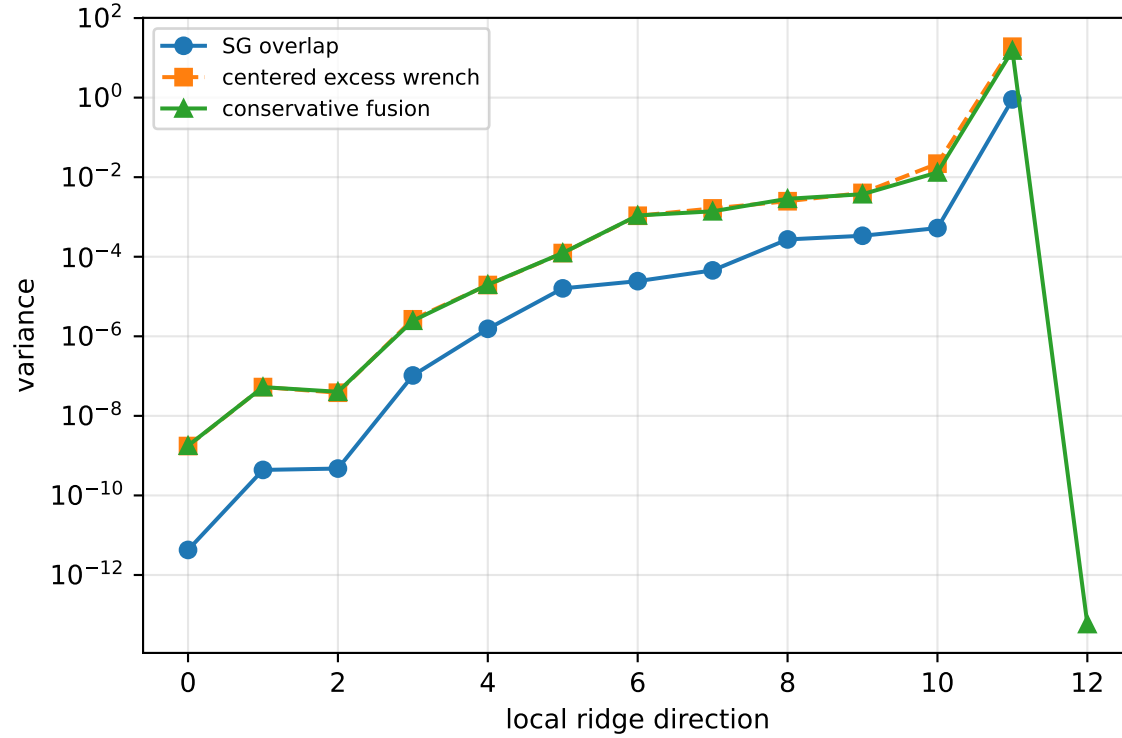
rotor lag cell: (0.152369261, 0.152392626] s

representative: 0.152380943 s

force_over_mass_rotor_4_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=7.48e-14)



force_over_mass_rotor_4_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 396.7$
wrench-to-acceleration recovery max error = $3.25\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

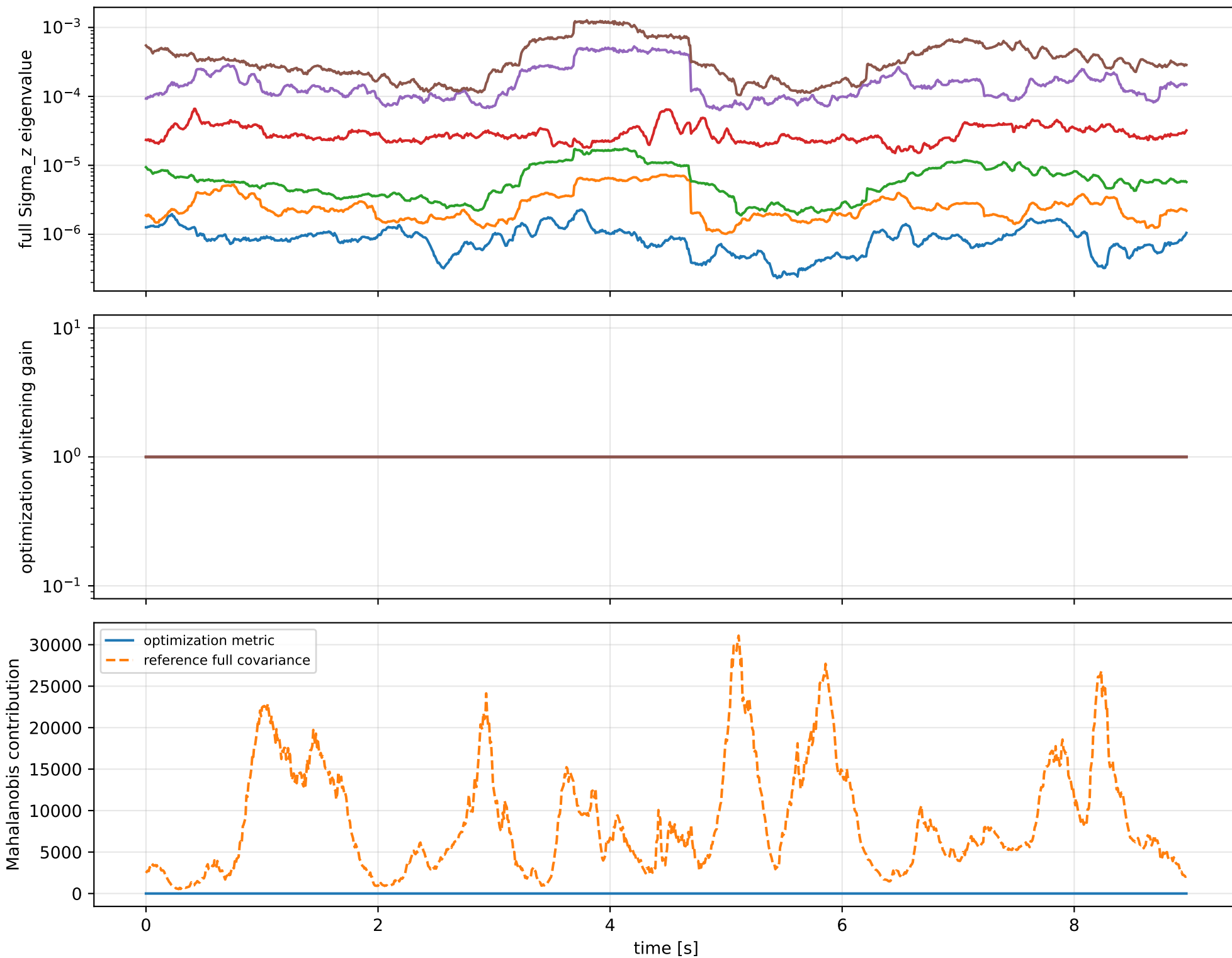
index 11: variance=15.26, ratio=16.94
 $\log_{\text{mass}} = -0.0283$; second_moment_diag_1=-0.311
second_moment_diag_2=+0.351; second_moment_diag_3=+0.0995
second_moment_offdiag_12=-0.818; second_moment_offdiag_13=+0.221
second_moment_offdiag_23=-0.217; cog_x=+0.000429
cog_y=-0.000439; cog_z=+0.000139
 $\log_{\text{force_eff_1}} = -0.0282$; $\log_{\text{force_eff_2}} = -0.0229$
 $\log_{\text{force_eff_3}} = -0.0284$; $\log_{\text{force_eff_4}} = -0.0312$

index 10: variance=0.01313, ratio=24.97
 $\log_{\text{mass}} = -0.11$; second_moment_diag_1=-0.00435
second_moment_diag_2=-0.246; second_moment_diag_3=+0.635
second_moment_offdiag_12=+0.207; second_moment_offdiag_13=+0.617
second_moment_offdiag_23=-0.198; cog_x=+0.0181
cog_y=-0.000548; cog_z=-0.0112
 $\log_{\text{force_eff_1}} = -0.168$; $\log_{\text{force_eff_2}} = +0.017$
 $\log_{\text{force_eff_3}} = +0.0477$; $\log_{\text{force_eff_4}} = -0.172$

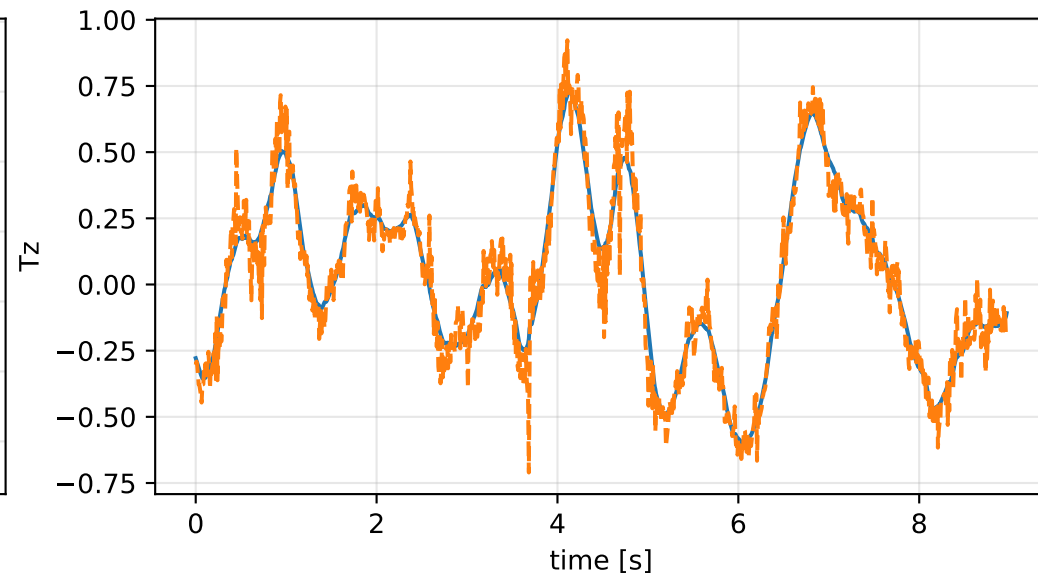
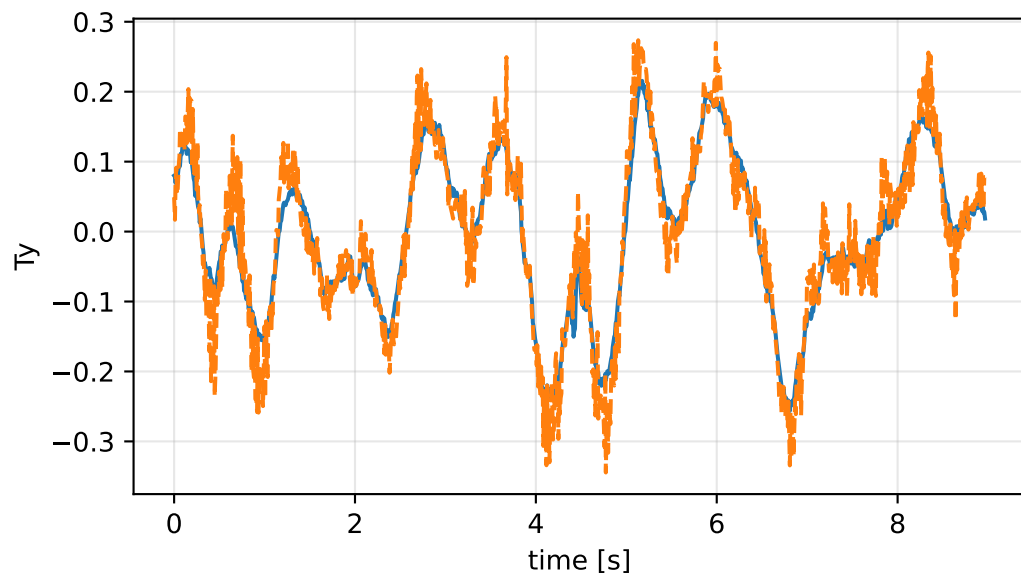
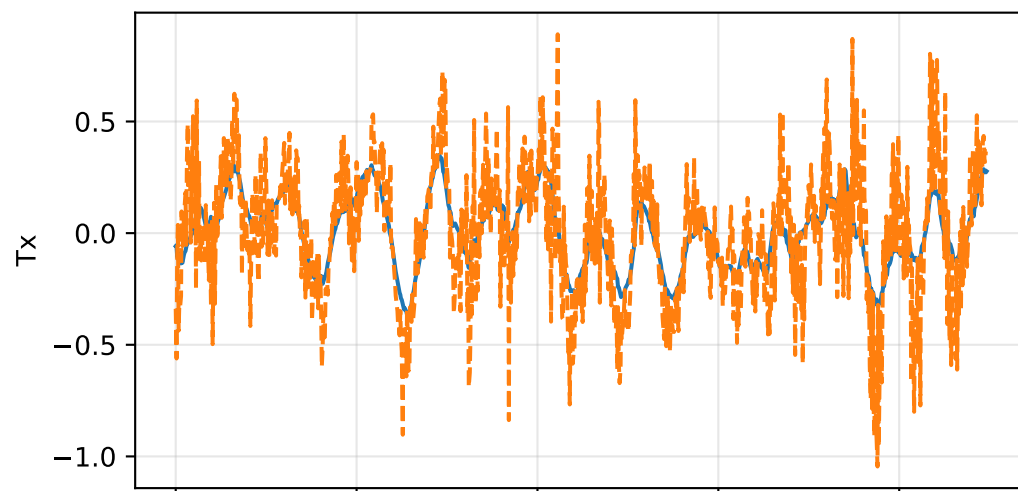
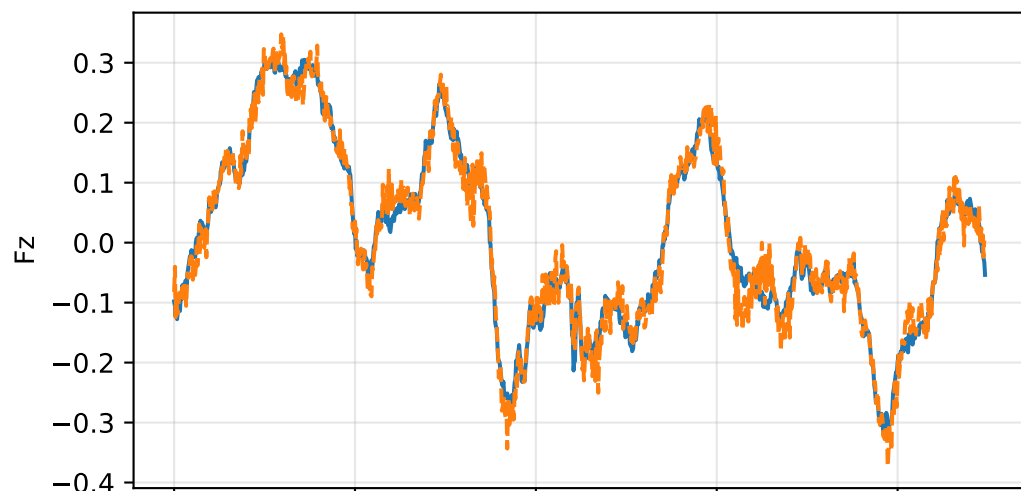
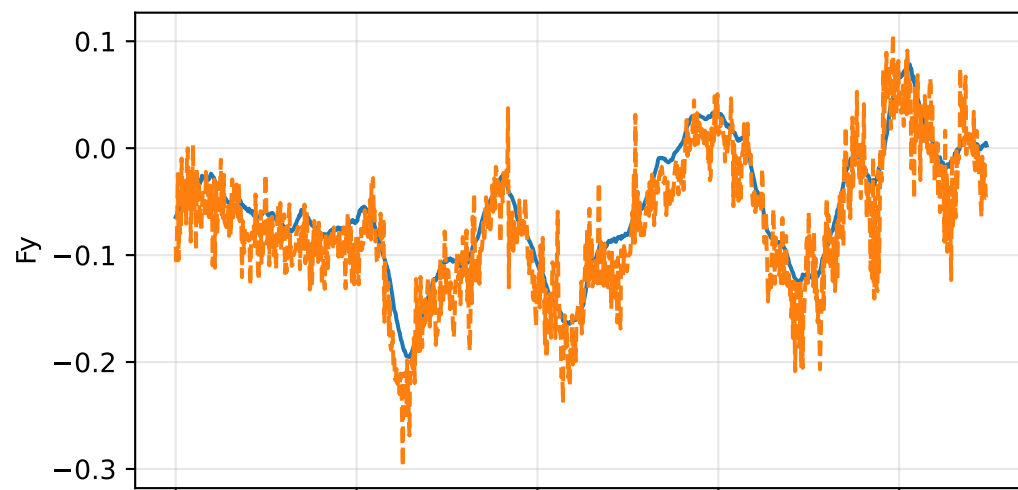
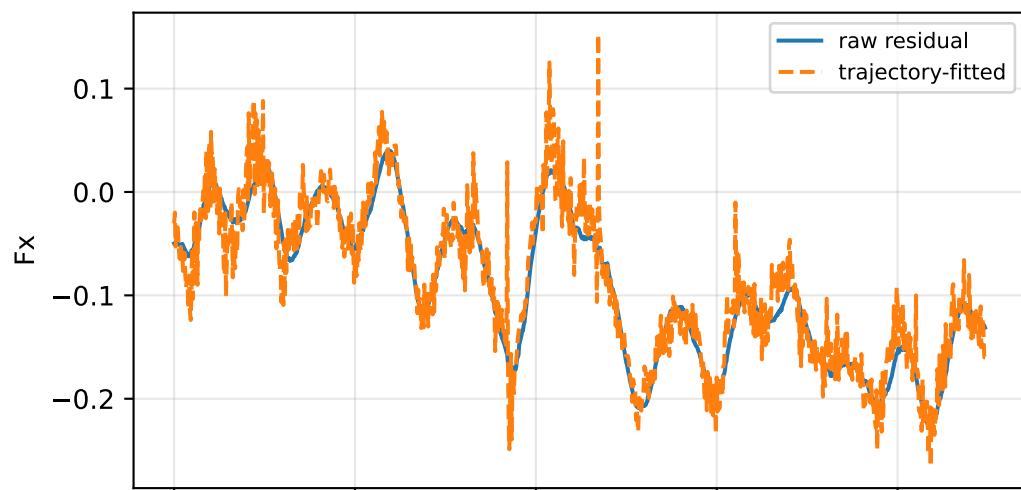
index 9: variance=0.003729, ratio=11.07
 $\log_{\text{mass}} = +0.0819$; second_moment_diag_1=-0.0116
second_moment_diag_2=-0.221; second_moment_diag_3=-0.451
second_moment_offdiag_12=+0.0174; second_moment_offdiag_13=+0.146
second_moment_offdiag_23=-0.536; cog_x=+0.047
cog_y=-0.0373; cog_z=+0.0361
 $\log_{\text{force_eff_1}} = +0.0544$; $\log_{\text{force_eff_2}} = +0.607$
 $\log_{\text{force_eff_3}} = +0.134$; $\log_{\text{force_eff_4}} = -0.194$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_4_nominal: optimization vs reference covariance



force_over_mass_rotor_4_nominal: wrench / closure (force max 6.74e-15, torque max 2.22e-16)



force_over_mass_rotor_4_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_4_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 678b0ccde06a50492269171f09bec596a62e0e49ae20754e771a2a2c2d4a9df2

data objective: 32.5928835216

prior objective: 1.88921840797e-08

total objective: 32.5928835405

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_4_nominal (force_effectiveness_over_mass)

components: rotor_4

target: [0.42524282]

std: [1.e-05]

final: [0.42524282]

error: [-1.94382016e-09]

standardized residual: [-0.00019438]

factor objective: 1.88921840797e-08